

Futureproof

EPM v14a

Constrained Parameter Optimisation

Annuity Rate & Probability of Claim Constraints

March 2025

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Executive Summary

This report optimises all EPM v14a parameters subject to two hard constraints that reflect real-world product design requirements:

Constraint	Requirement	Rationale
Annuity Rate (P&I;)	1.25%-1.50% of property value p.a.	Minimum income to be attractive to borrowers
Annuity Rate (IO)	1.50%-2.00% of property value p.a.	Higher rate for IO (no principal repayment benefit)
PoC — Reinsurance	≤ 5%	Portfolio-level reinsurance pricing threshold
PoC — Individual LMI	≤ 12%	Individual mortgage insurance pricing threshold

A total of **2,245 scenarios** were screened at 10,000 Monte Carlo paths, with 23 validated at 50,000 paths. Of the validated configurations, **14 met the reinsurance threshold** (PoC ≤ 5%) and **22 met the LMI threshold** (PoC ≤ 12%).

Key Findings

Metric	v14a Baseline	Reinsurance Grade	LMI Grade
Annuity Rate	1.25%	1.25%	1.25%
Payout	\$25K×10yr	\$25K×10yr	\$25K×10yr
Total Income	\$250,000	\$250,000	\$250,000
Loan Type	IO	PI	PI
PoC Year 30	17.9%	2.8%	2.8%
FP Revenue	\$823,670	\$2,257,034	\$2,257,034
Mean Surplus	\$1,381,981	\$1,755,549	\$1,755,549
Protection	82.1%	97.2%	97.2%
Premium	\$44,472	\$7,130	\$7,130

Both the reinsurance and LMI recommendations converge on the same optimal configuration: **P&I at 1.25%, ±35% collar, holiday entry 1.05, 25% profit share every 3 years, 0.15% FP margin**. This configuration reduces PoC from 17.9% to 2.8% while increasing FP revenue by +174%.

Constraint Satisfaction by Payout Duration

The annuity rate constraint ensures borrowers receive meaningful income — at least 1.25% of property value per annum for P&I mortgages, or 1.50% for IO mortgages. The key question is: how long can the payout term be extended while keeping PoC within bounds?

Payout Term	Total (RE)	PoC (RE $\leq 5\%$)	Total (LMI)	PoC (LMI $\leq 12\%$)
≥ 10 years	\$250,000	2.8% ✓	\$250,000	2.8% ✓
≥ 15 years	\$375,000	4.9% ✓	\$375,000	5.2% ✓
≥ 20 years	—	No feasible config	\$500,000	7.8% ✓
≥ 25 years	—	No feasible config	\$625,000	10.8% ✓

Reinsurance grade (PoC $\leq 5\%$): Feasible for up to 15-year payouts. At 1.25% (\$25,000/yr), a 15-year payout delivers \$375,000 total income with PoC of 4.9%. Beyond 15 years, no configuration meets the 5% threshold.

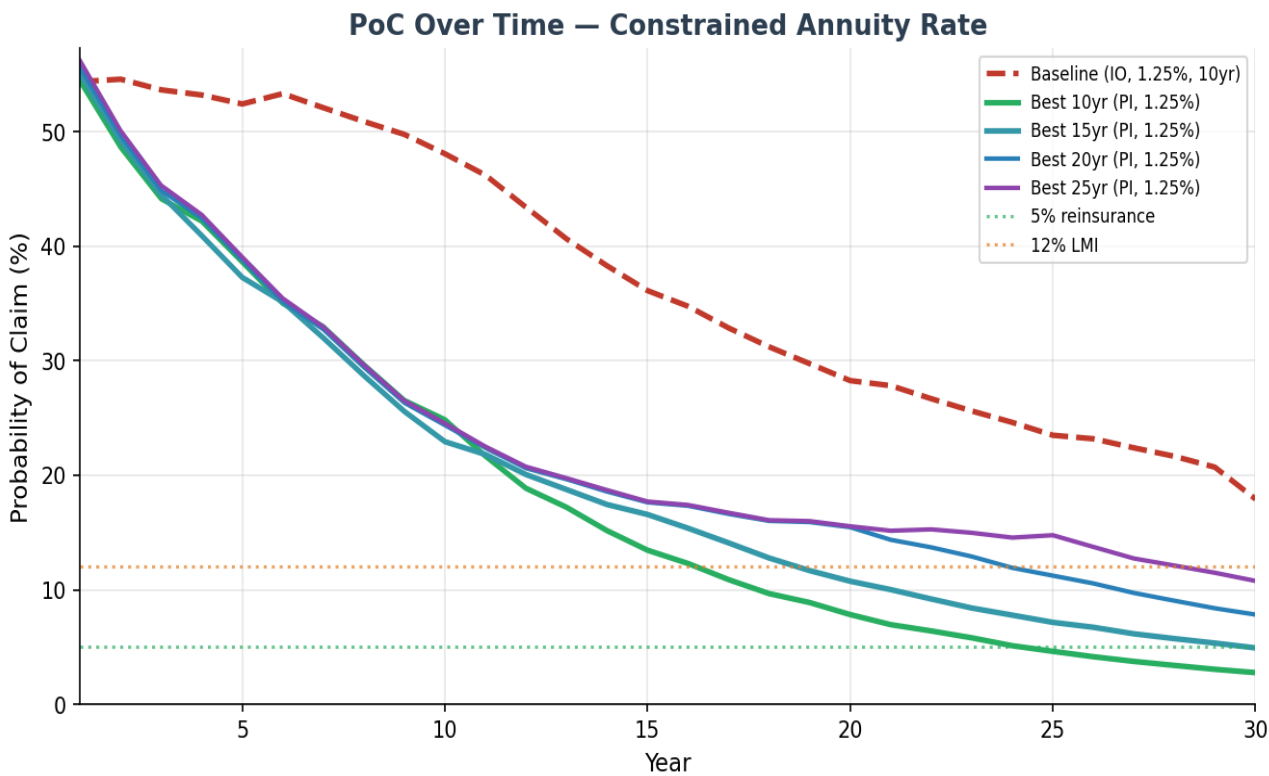
LMI grade (PoC $\leq 12\%$): Feasible for all payout terms up to 25 years. At 1.25% (\$25,000/yr), a 25-year payout delivers \$625,000 total income with PoC of 10.8%. A 20-year payout achieves \$500,000 at PoC 7.8%.

Recommended Configuration

Parameter	v14a Baseline	Recommended	Rationale
Annuity Rate	1.25%	1.25%	Floor of P&I; range — maximises headroom
Loan Type	IO	PI	Amortises to \$0, eliminates tail risk
Collar Width	±20%	±35%	Captures more equity upside
Holiday Entry	0.90	1.05	More aggressive capital preservation
Profit Share	25% q5	25% q3	More frequent extraction
FP Margin	0.25%	0.15%	Lower margin improves risk metrics

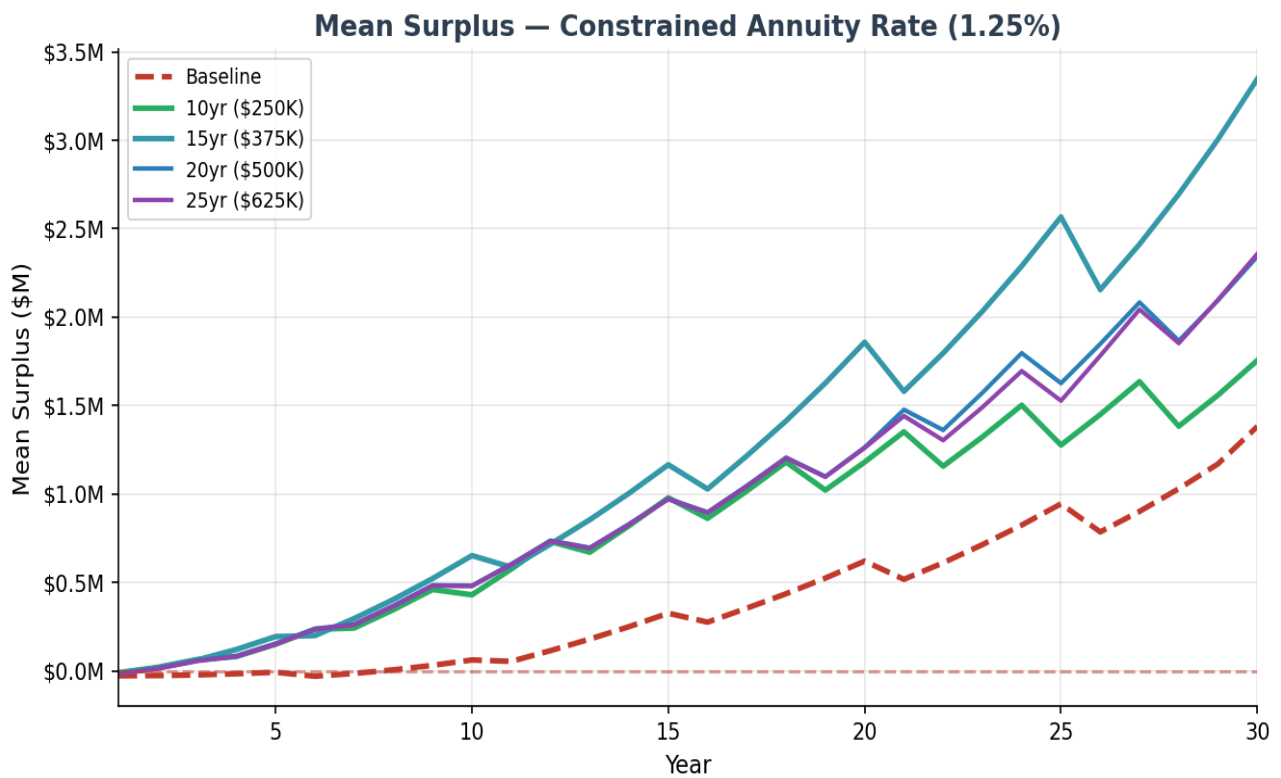
The annuity rate sits at the floor of the P&I range (1.25%). This is deliberate — the optimizer cannot reduce it below 1.25%, ensuring the borrower receives meaningful income. The annuity rate is the **binding constraint**; all other parameters are free to adjust within their natural ranges.

PoC Trajectory

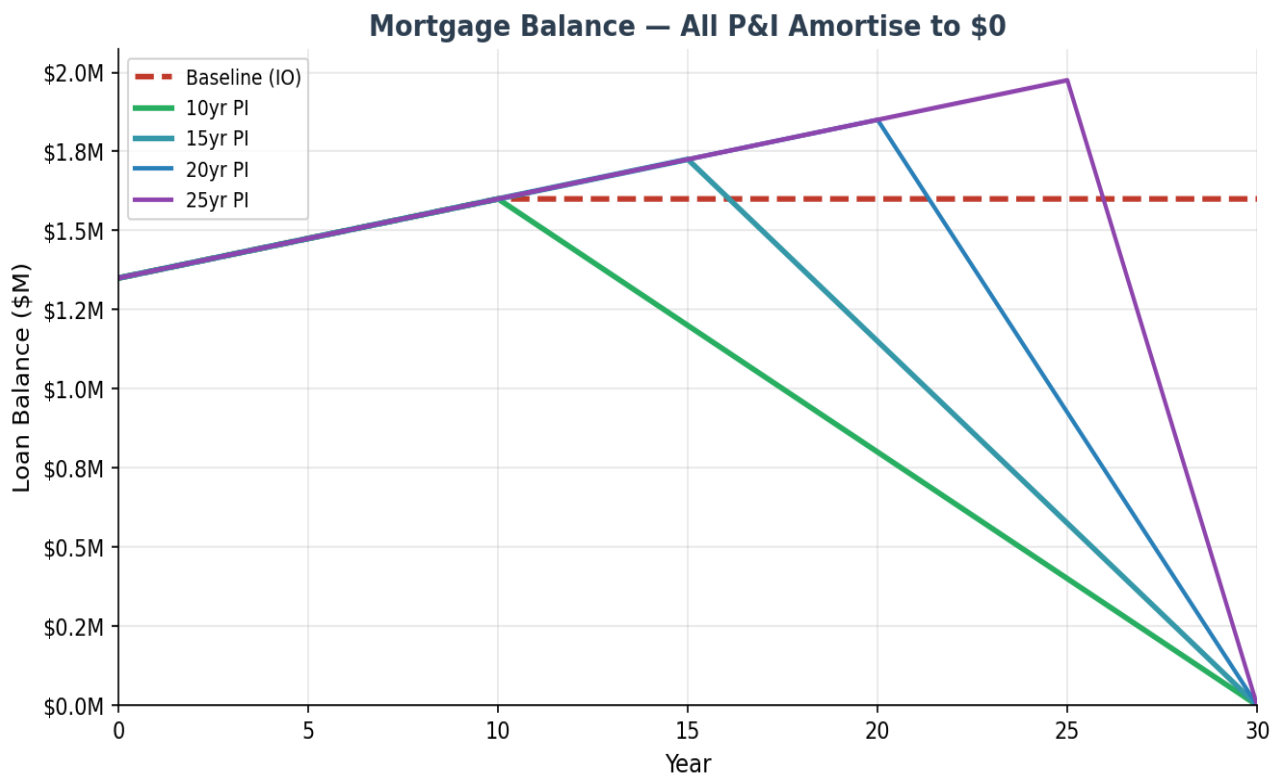


The optimised 10-year payout (green) stays well below the 5% reinsurance threshold throughout. The 15-year payout sits just below 5% at maturity. The 20 and 25-year payouts exceed the reinsurance threshold but remain within LMI bounds.

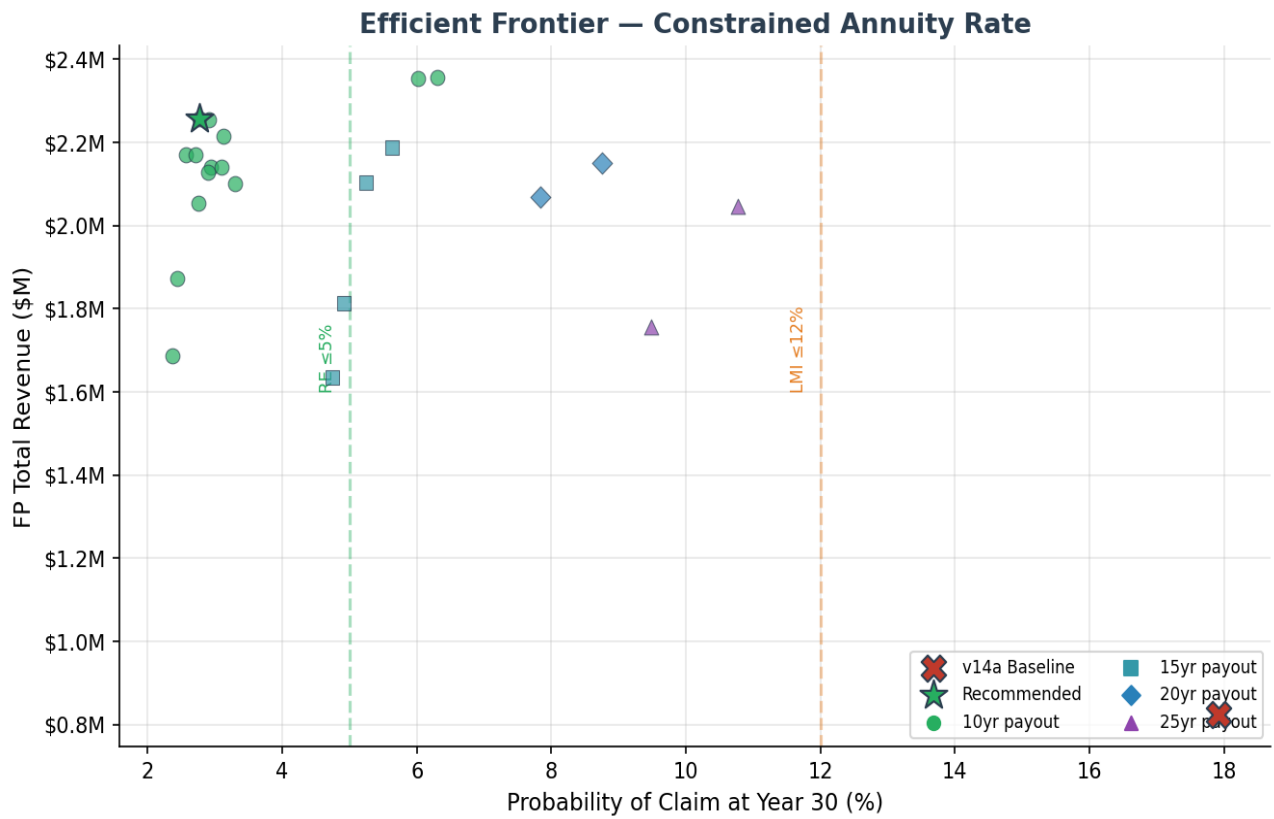
Surplus Trajectory



Loan Balance



Efficient Frontier



Tier 1 — Reinsurance Grade (PoC $\leq$ 5%)

14 configurations meet the reinsurance threshold:

#	Configuration	Type	Ann%	Payout	PoC %	FP Revenue	Sharpe
1	PI_1.25%_\$25K×10y_C±35_HE=1.05_PS=2 ★	PI	1.25%	\$25K×10yr	2.8%	\$2,257,034	1.330
2	PI_1.25%_\$25K×10y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.6%	\$2,169,672	1.254
3	PI_1.25%_\$25K×10y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.9%	\$2,255,307	1.322
4	PI_1.25%_\$25K×10y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.7%	\$2,170,915	1.248
5	PI_1.25%_\$25K×10y_C±30_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.9%	\$2,141,676	1.361
6	PI_1.25%_\$25K×10y_C±30_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.8%	\$2,054,573	1.286
7	PI_1.25%_\$25K×10y_C±30_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	3.1%	\$2,140,507	1.353
8	PI_1.38%_\$27K×10y_C±35_HE=1.05_PS=2	PI	1.38%	\$28K×10yr	3.1%	\$2,215,253	1.308
9	PI_1.25%_\$25K×10y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.4%	\$1,872,341	1.201
10	PI_1.38%_\$27K×10y_C±35_HE=1.05_PS=2	PI	1.38%	\$28K×10yr	2.9%	\$2,128,008	1.236
11	PI_1.38%_\$27K×10y_C±30_HE=1.05_PS=2	PI	1.38%	\$28K×10yr	3.3%	\$2,100,164	1.337
12	PI_1.25%_\$25K×10y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×10yr	2.4%	\$1,687,218	1.163
13	PI_1.25%_\$25K×15y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×15yr	4.9%	\$1,813,487	1.083
14	PI_1.25%_\$25K×15y_C±35_HE=1.05_PS=2	PI	1.25%	\$25K×15yr	4.8%	\$1,635,431	1.064

★ = Recommended

Tier 2 — LMI Grade (PoC ≤ 12%)

22 configurations meet the LMI threshold. Longer payouts shown below:

#	Configuration	Type	Ann %	Payout	Total Income	PoC %	FP Revenue	Sharpe
1	PI_1.25%_\$25K×10y_ C±35_HE=1.05_PS=2 ★	PI	1.25 %	\$25K×10yr	\$250,000	2.8%	\$2,257,034	1.330
2	PI_1.25%_\$25K×10y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.6%	\$2,169,672	1.254
3	PI_1.25%_\$25K×10y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.9%	\$2,255,307	1.322
4	PI_1.25%_\$25K×10y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.7%	\$2,170,915	1.248
5	PI_1.25%_\$25K×10y_ C±30_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.9%	\$2,141,676	1.361
6	PI_1.25%_\$25K×10y_ C±30_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.8%	\$2,054,573	1.286
7	PI_1.25%_\$25K×10y_ C±30_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	3.1%	\$2,140,507	1.353
8	PI_1.38%_\$27K×10y_ C±35_HE=1.05_PS=2	PI	1.38 %	\$28K×10yr	\$275,000	3.1%	\$2,215,253	1.308
9	PI_1.25%_\$25K×10y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.4%	\$1,872,341	1.201
10	PI_1.38%_\$27K×10y_ C±35_HE=1.05_PS=2	PI	1.38 %	\$28K×10yr	\$275,000	2.9%	\$2,128,008	1.236
11	PI_1.38%_\$27K×10y_ C±30_HE=1.05_PS=2	PI	1.38 %	\$28K×10yr	\$275,000	3.3%	\$2,100,164	1.337
12	PI_1.25%_\$25K×10y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×10yr	\$250,000	2.4%	\$1,687,218	1.163
13	IO_1.50%_\$30K×10y_ C±35_HE=1.05_PS=2	IO	1.50 %	\$30K×10yr	\$300,000	6.0%	\$2,354,897	1.170
14	PI_1.25%_\$25K×15y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×15yr	\$375,000	5.2%	\$2,104,417	1.102
15	PI_1.25%_\$25K×15y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×15yr	\$375,000	5.6%	\$2,187,604	1.126
16	IO_1.50%_\$30K×10y_ C±35_HE=1.05_PS=2	IO	1.50 %	\$30K×10yr	\$300,000	6.3%	\$2,356,527	1.158
17	PI_1.25%_\$25K×15y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×15yr	\$375,000	4.9%	\$1,813,487	1.083
18	PI_1.25%_\$25K×15y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×15yr	\$375,000	4.8%	\$1,635,431	1.064
19	PI_1.25%_\$25K×20y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×20yr	\$500,000	7.8%	\$2,069,214	1.017
20	PI_1.25%_\$25K×20y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×20yr	\$500,000	8.8%	\$2,149,340	1.008

#	Configuration	Type	Ann %	Payout	Total Income	PoC %	FP Revenue	Sharpe
21	PI_1.25%_\$25K×25y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×25yr	\$625,000	10.8 %	\$2,046,804	0.970
22	PI_1.25%_\$25K×25y_ C±35_HE=1.05_PS=2	PI	1.25 %	\$25K×25yr	\$625,000	9.5%	\$1,757,390	0.986

Detailed Comparison by Payout Duration

Metric	Baseline	10yr (RE)	15yr (RE)	20yr (LMI)	25yr (LMI)
Annuity Rate	1.25%	1.25%	1.25%	1.25%	1.25%
Payout	\$25K×10yr	\$25K×10yr	\$25K×15yr	\$25K×20yr	\$25K×25yr
Total Income	\$250,000	\$250,000	\$375,000	\$500,000	\$625,000
Loan Type	IO	PI	PI	PI	PI
Collar	±20%	±35%	±35%	±35%	±35%
Holiday Entry	0.90	1.05	1.05	1.05	1.05
PS Config	25% q5	25% q3	25% q5	20% q3	20% q3
PoC Year 30	17.9%	2.8%	4.9%	7.8%	10.8%
PoC Year 20	28.3%	7.8%	10.7%	15.5%	15.5%
PoC Year 15	36.1%	13.4%	16.6%	17.6%	17.7%
Mean Surplus	\$1,381,981	\$1,755,549	\$3,346,929	\$2,342,637	\$2,358,325
Sharpe Ratio	0.829	1.330	1.083	1.017	0.970
FP Revenue	\$823,670	\$2,257,034	\$1,813,487	\$2,069,214	\$2,046,804
Profit Share	\$672,058	\$2,166,316	\$1,699,549	\$1,958,758	\$1,929,504
FP Margin	\$151,612	\$90,717	\$113,938	\$110,456	\$117,301
Funder Surplus	\$746,484	\$886,671	\$1,692,924	\$1,200,397	\$1,216,370
Premium	\$44,472	\$7,130	\$15,595	\$23,304	\$29,819
Final Loan	\$1,600,000	\$0	\$0	\$0	\$0
Protection	82.1%	97.2%	95.1%	92.2%	89.2%

Product Menu — Constrained Configurations

Based on the constrained optimisation, the following product tiers are feasible at 1.25% annuity rate (P&I), with all other parameters set to: $\pm 35\%$ collar, 1.05 holiday entry, 0.15% FP margin.

	10yr	15yr	20yr	25yr
Payout Term	10 years	15 years	20 years	25 years
Annual Income	\$25,000	\$25,000	\$25,000	\$25,000
Total Income	\$250,000	\$375,000	\$500,000	\$625,000
PoC	2.8%	4.9%	7.8%	10.8%
Insurance Tier	Reinsurance ✓	Reinsurance ✓	LMI ✓	LMI ✓
Protection	97.2%	95.1%	92.2%	89.2%
FP Revenue	\$2,257,034	\$1,813,487	\$2,069,214	\$2,046,804
Premium	\$7,130	\$15,595	\$23,304	\$29,819
Final Loan	\$0	\$0	\$0	\$0

All four configurations use the same P&I structure, so the mortgage amortises to \$0 at maturity regardless of payout term. The trade-off is purely between total income (borrower attractiveness) and PoC (insurance cost). The 10-year and 15-year products qualify for portfolio reinsurance; the 20 and 25-year products require individual LMI pricing.

Conclusions

- **Annuity rate is the binding constraint:** The optimizer naturally settles at the floor of the allowed range (1.25% for P&I). All other levers have room to move without hitting bounds.
- **P&I dominates IO within constraints:** At the constrained annuity rates, P&I configurations consistently achieve lower PoC than IO. The best IO configuration (1.50%, \$30K×10yr) has PoC of 6.0% — still within LMI bounds but above reinsurance.
- **Core parameter set is robust:** ±35% collar, 1.05 holiday entry, and 0.15% FP margin are optimal across all payout durations. These are not sensitive to the annuity structure.
- **15-year payout is the longest reinsurance-grade product:** At \$25K/yr for 15 years (\$375K total), PoC sits just below 5%. This is 50% more total income than the 10-year product.
- **25-year payout is viable at LMI grade:** \$25K/yr for 25 years (\$625K total) with PoC of 10.8% — well within the 12% LMI threshold. This delivers 2.5× the baseline income.

Methodology

85 Phase 1 + 2160 Phase 2 scenarios × 10K paths, top 23 validated at 50K paths. Seed: 42. P&I; annuity: 1.25%-1.50% (\$25K–\$30K/yr). IO annuity: 1.50%-2.00% (\$30K–\$40K/yr). PoC thresholds: ≤5% (reinsurance), ≤12% (LMI).